

Mathematical Foundations of Machine Learning

Midterm 2

Time: 90 minutes

Total: 100 points

Question 1 (15 points)

Given that $\lambda = -2$ is an eigenvalue of

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & 0 \\ -4 & -1 & a \\ b & 2 & -2 \end{bmatrix},$$

and that $\boldsymbol{\alpha} = \begin{bmatrix} 1 \\ c \\ -2 \end{bmatrix}$ is an eigenvector of $\mathbf{A}^{-1}$ corresponding to eigenvalue $\lambda_0 = -\frac{1}{2}$, find the values of a , b , and c .

Question 2 (15 points)

(a) Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 4 & -2 & 0 \\ 0 & 2 & 0 \\ 2 & -2 & 2 \end{bmatrix}.$$

Compute its eigendecomposition, i.e., find matrices $\mathbf{P}$ and $\mathbf{D}$ in $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$.

(b) Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

Compute its singular value decomposition (SVD), i.e., $\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, where $\mathbf{U} \in \mathbb{R}^{3 \times 3}$ is orthogonal, $\boldsymbol{\Sigma} \in \mathbb{R}^{3 \times 2}$ is with non-negative diagonal entries, and $\mathbf{V} \in \mathbb{R}^{2 \times 2}$ is orthogonal.

Question 3 (15 points)

If $\mathbf{A} \in \mathbb{R}^{n \times n}$ with SVD $\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ for singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$, left singular vectors $\mathbf{u}_i$, and right singular vectors $\mathbf{v}_i$, then find the largest constant $c_1 \in \mathbb{R}$ and smallest constant $c_2 \in \mathbb{R}$ so that the following inequality holds for all vectors $\mathbf{x} \in \mathbb{R}^n$:

$$c_1 \|\mathbf{x}\| \leq \|\mathbf{A}\mathbf{x}\| \leq c_2 \|\mathbf{x}\|.$$

Give one vector $\mathbf{x}$ that attains the lower bound and one vector $\mathbf{x}$ that attains the upper bound.

Question 4 (15 points)

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The **Frobenius norm** of $\mathbf{A}$ is defined as the square root of the trace of $\mathbf{A}^\top \mathbf{A}$:

$$\|\mathbf{A}\|_F = \sqrt{\text{Tr}(\mathbf{A}^\top \mathbf{A})}.$$

(a) Prove that the norm is equivalent to the sum of squared singular values of $\mathbf{A}$:

$$\|\mathbf{A}\|_F = \sqrt{\text{Tr}(\mathbf{A}^\top \mathbf{A})} = \sqrt{\sum_i \sigma_i^2(\mathbf{A})}.$$

(b) Prove that for any orthogonal matrices $\mathbf{U} \in \mathbb{R}^{m \times m}$ and $\mathbf{V} \in \mathbb{R}^{n \times n}$, the Frobenius norm is orthogonally invariant:

$$\|\mathbf{U}\mathbf{A}\mathbf{V}\|_F = \|\mathbf{A}\|_F.$$

Hint: The cyclic property of the trace: $\text{Tr}(\mathbf{XYZ}) = \text{Tr}(\mathbf{ZXY})$ for matrices of compatible dimensions.

Question 5 (10 points)

Let $\mathbf{g} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and $\mathbf{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined as

$$\mathbf{g}(s, t) = \begin{bmatrix} s^2 t \\ st^2 \\ s + t \end{bmatrix}, \quad \mathbf{f}(\mathbf{x}) = \begin{bmatrix} x_1 x_2 + x_3^2 \\ e^{x_1 + x_2} \end{bmatrix}.$$

Compute the Jacobian matrix of the composite function $\mathbf{h} = \mathbf{f} \circ \mathbf{g} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ at the point $(s, t) = (1, 1)$, i.e. find $\frac{d\mathbf{h}}{d(s, t)}$ for $(s, t) = (1, 1)$.

Question 6 (15 points)

Consider the regularized loss function

$$L(\mathbf{W}) = \|\mathbf{y} - \mathbf{W}\mathbf{x}\|^2 + \lambda \text{Tr}(\mathbf{W}^\top \mathbf{W}),$$

where $\mathbf{W} \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^m$, and scalar $\lambda > 0$. Compute the gradient $\partial L / \partial \mathbf{W}$ and write down the normal equation obtained by setting the gradient to zero.

Question 7 (15 points)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = \ln(1 + e^{ax+by})$, where $a, b \in \mathbb{R}$ are constants. At the point $(x_0, y_0) = (0, 0)$:

- (1) Compute the gradient $\nabla_{x,y} f(0, 0)$.
- (2) Compute the Hessian matrix $\mathbf{H}(0, 0)$.
- (3) Write the second-order Taylor expansion $T_2(x, y)$ of f at $(0, 0)$.